Route G-R10: All-Channel Readout and an Exact Stationary Conditional Clock

Unchanged vertex contact; explicit preparation and measurement limits

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Abstract

The extra resonant channel found in G-R9 supports, rather than prevents, an exact stationary dark subspace. Within it a subsystem's phase reading labels another subsystem's state without the auxiliary update index. All nine states and all readout outcomes are retained. However, the original exchange input has only 2/3 dark weight, conserved by this contact: existence is not autonomous preparation. Generic stationary interacting states also need not follow the bare-gap rigid clock orbit. This is a finite conditional quantum construction, not a derivation of physical seconds, records, an arrow, spacetime or particle masses.

1 Same contact and a complete observation algebra

Keep G-R9's two independent three-level copies. At its already declared resonance $\theta = \pi/2$, in the fixed Fourier-mode basis,

$$L = \lambda_0 I + \Delta N, \quad N = \mathrm{diag}(0,1,2), \quad \lambda_0 = 2 - \sqrt{3}, \quad \Delta = \sqrt{3}. \tag{1}$$

The full dimensionless generator is unchanged:

$$H = L \otimes I + I \otimes L + \gamma P_v \otimes P_v, \quad v = \frac{(1,1,1)^T}{\sqrt{3}}, \quad P_v = vv^\dagger. \tag{2}$$

Here P_v is one vertex projector expressed in mode coordinates; no outlet is removed. $\gamma = 0.09, 0.03, 0.01$ are inherited fixtures, not fits.

The Hermitian local basis consists of $Z_k = |k\rangle\langle k|$ and, for $k < l$,

$$X_{kl} = |k\rangle\langle l| + |l\rangle\langle k|, \qquad Y_{kl} = -\mathrm{i}|k\rangle\langle l| + \mathrm{i}|l\rangle\langle k|. \tag{3}$$

These nine matrices span the local algebra; their 81 products span the joint algebra. Incompatible settings require separate runs. For the clock readout fix seven effects before choosing a state:

$$\phi_r = 2\pi r/7, \quad w(\phi) = (1, e^{\mathrm{i}\phi}, e^{2\mathrm{i}\phi})^T, \quad E_r = \frac{w(\phi_r)w(\phi_r)^\dagger}{7}, \quad \sum_{r=0}^{6} E_r = I. \tag{4}$$

Completeness follows from the finite Fourier sum for differences $k - l \in \{-2, \ldots, 2\}$. Positivity is immediate. The phase effects alone span only five local dimensions: this readout is not tomography.

2 Joint probabilities, unobserved outcomes and a clock criterion

For any full nine-state density matrix, define

$$p_{rs} = \mathrm{tr}[\rho(E_r \otimes E_s)], \qquad R_B(r) = \mathrm{tr}_A[(E_r \otimes I)\rho], \tag{5}$$

$$p_r = \mathrm{tr}\, R_B(r), \qquad \rho_{B|r} = R_B(r)/p_r. \tag{6}$$

For $p_r > 0$ this is the remote conditional state for any local instrument on A with effect E_r; cyclicity of the partial trace removes its particular Kraus representation. It says nothing about a later A measurement or the apparatus energy. Without an A record,

$$\sum_r p_r \rho_{B|r} = \mathrm{tr}_A\, \rho. \tag{7}$$

No preferred exit is postselected. The continuous extension has effects $w(\phi)w(\phi)^\dagger\, d\phi/(2\pi)$; it is a POVM, not an infinity of orthogonal timestamps.

For the resonant triple, write

$$|\psi\rangle = c_0|0,2\rangle + c_1|1,1\rangle + c_2|2,0\rangle, \quad \sum_k |c_k|^2 = 1. \tag{8}$$

All phase marginals are uniform, $p_r = 1/7$, and the normalized B vector is

$$|\psi_B(\phi)\rangle = \sum_{k=0}^{2} c_k e^{-\mathrm{i}k\phi}|2-k\rangle = e^{-2\mathrm{i}\phi} e^{\mathrm{i}N_B\phi}|\psi_B(0)\rangle. \tag{9}$$

Thus $\rho_{B|\phi} = e^{\mathrm{i}N_B\phi}\rho_{B|0}e^{-\mathrm{i}N_B\phi}$. The same conclusion holds for mixed states supported on this triple. This is an exact conditional orbit, but a generic triple state is not stationary under the full contact. The next section identifies states that satisfy both requirements.

For the weak-contact exchange approximation from $|1,1\rangle$, $c_0 = c_2 = a = (z-1)/3$, $c_1 = b = (z+2)/3$, $z = e^{-\mathrm{i}x}$, $x = \gamma s/3$, $s = \alpha n$, $\alpha = 1/4$. All three amplitudes give

$$p_{rs} = \frac{|b + 2a\cos(\phi_r - \phi_s)|^2}{49}. \tag{10}$$

The relative-phase harmonics are $C_1 = -|a|^2$ and $C_2 = |a|^2$. Correlations change with the assumed exchange index through a, b; Eq. (9) alone has not eliminated this preparation dependence.

We test the full contact against this *specified bare-gap* orbit:

$$D_7(\rho) = \max_{r=0,\ldots,6} \frac{1}{2} \left\| \rho_{B|r} - e^{\mathrm{i}N_B\phi_r}\rho_{B|0}e^{-\mathrm{i}N_B\phi_r} \right\|_1. \tag{11}$$

A nonzero value rejects that rigid description on the actual seven readings, not every possible interacting-clock interpretation.

3 An exact dark clock, and an exact preparation obstruction

The triple's free total eigenvalue is 4. The contact has every mode-basis element $\gamma/9$, so a triple vector is annihilated iff $c_0 + c_1 + c_2 = 0$. An orthonormal basis is

$$|D_+\rangle = \frac{|0,2\rangle - 2|1,1\rangle + |2,0\rangle}{\sqrt{6}}, \qquad |D_-\rangle = \frac{|0,2\rangle - |2,0\rangle}{\sqrt{2}}. \tag{12}$$

For every γ in Eq. (2), not just a weak-coupling limit,

$$V|D_\pm\rangle = 0, \qquad H|D_\pm\rangle = 4|D_\pm\rangle. \tag{13}$$

Consequently every density matrix supported on this dark subspace is stationary and has the conditional orbit in Eq. (9). No update number occurs in that readout relation. Defining a formal dial orientation $\xi = -\phi$ gives $\mathrm{i}\partial_\xi \rho_{B|\xi} = [N_B, \rho_{B|\xi}]$. Its unit slope follows from the chosen equal-gap spectrum; it is not a prediction of a universal physical time rate.

For D_+ the continuous relative-phase density, $\delta = \phi_A - \phi_B$, is

$$p(\delta) = \frac{(1-\cos\delta)^2}{3\pi}, \qquad C_1 = -\frac{2}{3}, \quad C_2 = \frac{1}{6}. \tag{14}$$

It is broad and peaks at a fixed offset $\delta = \pi$; equal phase outcomes have zero joint probability. Ignoring A gives diagonal $\rho_B = \mathrm{diag}(1,4,1)/6$ and a uniform B phase distribution. Local mode dephasing preserves all energy populations and both reduced states but makes the joint phase table flat. This is a coherence control, not a unique experimental entanglement witness. The dephased control need not itself remain stationary under H.

D_- instead gives $p(\delta) = \sin^2\delta/\pi$, with a twofold ambiguity in the continuous dial. Identical copies keep D_- an exact eigenstate even off the equal-gap resonance: the two exchanged levels always have equal total energy. D_+ loses stationarity off resonance. This robustness tradeoff must not be concealed by retuning a clock.

Let $P_D = |D_+\rangle\langle D_+| + |D_-\rangle\langle D_-|$. The full contact obeys

$$[H, P_D] = 0, \qquad \langle 1,1|P_D|1,1\rangle = \frac{2}{3}, \qquad |\langle D_+|e^{-\mathrm{i}\alpha n H}|1,1\rangle|^2 = \frac{2}{3}. \tag{15}$$

The antisymmetric overlap remains zero. Thus the original unitary exchange cannot prepare a pure dark clock. A hypothetical dark-energy filter would succeed with probability 2/3 and reject 1/3; its instrument, energy accounting and records would be new inputs. The dark clock also has zero contact action: persistent correlations are not proof of ongoing energy exchange or clock synchronization.

4 Full stationary readout and the retained interaction term

For a normalized full eigenstate $H|\Psi\rangle = E|\Psi\rangle$, set $q_\phi = w(\phi)/\sqrt{3}$ and the unnormalized conditional amplitude $\chi(\phi) = (\langle q_\phi| \otimes I)|\Psi\rangle$. Since $\langle q_\phi|N_A|\Psi\rangle = \mathrm{i}\partial_\phi\chi$ and $q_0 = v$,

$$\mathrm{i}\Delta\partial_\phi\chi(\phi) = (E - 2\lambda_0 - \Delta N_B)\chi(\phi) - \gamma f(\phi)P_v\chi(0), \quad f(\phi) = \frac{1 + e^{-\mathrm{i}\phi} + e^{-2\mathrm{i}\phi}}{3}. \tag{16}$$

This is exact on all nine states. The contact couples different dial readings, rather than merely rescaling time. For a dark state $P_v\chi(0) = 0$ and $E - 2\lambda_0 = 2\Delta$, yielding Eq. (9). For generic eigenstates the term survives and $\|\chi(\phi)\|$ can vary. Normalizing it without retaining that likelihood would give a wrong conditional equation. Conditional dynamics from stationary correlations is established Page–Wootters reasoning [1]; interactions can produce nonlocal clock-label equations [2]. The specific finite contact kernel above is derived here, not a new general mechanism.

An unread update is a specified prior, not an automatic law. Let $U = e^{-\mathrm{i}\alpha H}$ and assume the uniform Cesaro average

$$\bar{\rho} = \lim_{T\to\infty} \frac{1}{T}\sum_{n=0}^{T-1} U^n\rho_0 U^{-n} = \sum_E \Pi_E\rho_0\Pi_E, \qquad \rho_0 = |1,1\rangle\langle 1,1|. \tag{17}$$

For the declared fixtures $\alpha(E_{\max} - E_{\min}) < 2\pi$, so there are no distinct-energy update-phase aliases. The equality follows by summing each off-diagonal geometric series. This prior, not an arbitrary unknown distribution of n, gives $[H, \bar{\rho}] = 0$. It need not give $[N_A + N_B, \bar{\rho}] = 0$ or $D_7 = 0$.

Stationary effects can also be defined without deleting outcomes:

$$\bar{F}_{rs} = \sum_E \Pi_E(E_r \otimes E_s)\Pi_E, \qquad \sum_{rs}\bar{F}_{rs} = I, \quad [H, \bar{F}_{rs}] = 0. \tag{18}$$

They are positive and give the same probabilities as $E_r \otimes E_s$ on stationary ρ. They are generally *joint*, not local product effects. This logical POVM does not establish an autonomous, energy-conserving, locally implemented recording apparatus.

Order of limits matters. Writing $|B\rangle = (|0,2\rangle + |1,1\rangle + |2,0\rangle)/\sqrt{3}$, the positive weak-coupling limit after Eq. (17) is

$$\bar{\rho}_{0^+} = \frac{2}{3}|D_+\rangle\langle D_+| + \frac{1}{3}|B\rangle\langle B|. \tag{19}$$

The bright eigenvector tends to B, the dark weight stays $2/3$, and other multiplet overlaps vanish. This mixed state has $(C_1, C_2) = (-2/9, 2/9)$ and a rigid conditional orbit. Setting $\gamma = 0$ *before* averaging instead leaves ρ_0, with flat phase statistics. Weak coupling is not instantaneous dark-state preparation.

5 Numerical checks and physical interpretation

Use the same seven effects for every state. No reference phase, graph or channel is refitted. At each inherited γ, the exchange state is evaluated at the integer n nearest G-R9's first secular transfer peak. The mixed state uses Eq. (17), retaining all energy blocks. Define $A_7 = \max_r(7p_r) - \min_r(7p_r)$ to expose conditional likelihoods.

γ	full nine-state input	D_7	A_7
0.09	exchange, $n = 419$	0.0277141	0.0232937
0.09	uniform unread updates	0.0108673	0.000433842
0.03	exchange, $n = 1257$	0.0116668	0.00652081
0.03	uniform unread updates	0.00359595	0.0000482796
0.01	exchange, $n = 3770$	0.00332006	0.00235708
0.01	uniform unread updates	0.00119550	0.00000536533
any declared γ	exact D_+	0	0

Zeros in the last row are analytic identities, not fitted numerical thresholds. Nonzero D_7 rejects only the fixed bare-gap orbit. A more general interacting conditional description retains Eq. (16); no universal impossibility of relational time is inferred.

Verification scope. The program is `verify_gr10_relational_readout.py`.
Fixed inputs are in `RELATIONAL_READOUT_CARD.json`. Its 151 checks cover observation-algebra ranks, positive complete effects, exact dark eigenvectors, all-outcome marginalization, pure and mixed conditional orbits, dephased controls, full site/mode equivalence, invariant effects, the contact kernel, the preparation obstruction and flux-detuning controls. The finite contact matrices and spectral projectors are calculated independently of the analytic triple formulae. Earlier G-R9 files are hash-pinned and unchanged. These checks are not data.

What is now possible. The missing channel supports an exact stationary conditional clock in the *same* local-contact model. Unlike the previous ordered-update example, its conditional readout contains no n. But quantum kinematics, the tensor split, phase effects, resonance and state preparation remain assumptions. The phase has finite resolution and is cyclic; no successive records or arrow have been derived. No statement about physical E_d, time dilation, a spatial metric or masses follows.

Next bounded decision. Either specify one preparation plus one-record instrument, including its energy, reference and discarded-outcome costs, or keep the unfiltered mixed stationary state and its genuinely interacting conditional equation. Do not silently add a dissipative bath to purify the conserved 2/3 dark weight. The former tests operational realization; the latter avoids a pure-state preparation claim. Neither requires more gravitational-guide calibration or a new mass fit.

References

[1] D. N. Page and W. K. Wootters, *Evolution without evolution: Dynamics described by stationary observables*, Phys. Rev. D 27, 2885 (1983).

[2] A. R. H. Smith and M. Ahmadi, *Quantizing time: Interacting clocks and systems*, Quantum 3, 160 (2019), arXiv:1712.00081.